Designing NeuroLink: A Mobile Tool for Traumatic Brain Injury Communication

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Abstract

Traumatic brain injury (TBI) presents complex challenges for both affected individuals and their care teams, which often include healthcare providers and informal caregivers. Individuals with TBI often experience comorbidities that negatively impact their quality of life, with fatigue and mood impairments posing significant barriers to effective communication. These impairments can lead to social withdrawal and reduced help-seeking behaviours. To address this gap, NeuroLink – a mobile health application (mHealth app) – was developed to support individuals with TBI in tracking mood and fatigue in real-time. NeuroLink features a low-effort, intuitive interface designed for users experiencing cognitive fatigue, while also enabling secure data sharing with caregivers and clinicians. By integrating AI-powered pattern recognition and communication-enhancing features, the app transforms invisible struggles into actionable insights – facilitating early intervention, improving patient-caregiver interactions, and personalizing recovery pathways. NeuroLink represents a patient-centred, technology-enabled approach to holistic TBI recovery by bringing emotional health and communication to the forefront of care. Given the growing role of digital health solutions, NeuroLink offers a daily-use tool that enhances quality of care by supporting fatigue and mood regulation and improving social communication throughout the recovery journey.

Keywords

traumatic brain injury, mobile health, caregiver communication, usability testing, cognitive accessibility

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1 Introduction

TBI is a sudden injury that can lead to long-term complications. It is typically categorized as mild, moderate, or severe, based on the extent and nature of damage to the brain [1]. Mild TBI is characterized by a Glasgow Coma Scale (GCS) score of 13-15 and has a mortality rate of 0.1%. Moderate TBI, with a GCS score of 9-12, carries a mortality rate of 10%. Severe TBI, defined by a GCS score of less than 9, has a significantly higher mortality rate of 40%. These severity levels are critical in determining the prognosis and care required for individuals affected by TBI [2]. Individuals affected by TBI are at an increased risk for long-term comorbidities; it was found that those affected by moderate-severe TBI face a higher likelihood of additional medical and mental health comorbidities throughout their recovery [3]. An important comorbidity that is often overlooked in the recovery of TBI patients, is fatigue and its resulting mood instability. Approximately 18% to 75% of individuals experience fatigue after a TBI, and secondary fatigue is a phenomena that directly correlates to mood disorders [4]. TBIs increase the risk of emotional and behavioural dysregulation, such as agitation and personality changes. Studies indicate that major depressive disorder affects approximately 53% of TBI patients [5]. These emotional challenges can severely impact rehabilitation and diminish quality of life. When navigating the fatigue and mood instability associated with TBIs, social communication becomes a barrier with receiving support from caregivers and healthcare providers. It is reported that younger and older adults with TBI have significantly more frequent difficulties with social communication compared to those of age-matched control groups. Notably, older adults with TBI also self-reported more frequent social communication challenges than their non-TBI peers. In contrast, younger adults with

TBI reported similar levels of difficulty as non-TBI younger adults [6].

In addition to the toll TBIs can take on affected patients, the impact of this condition also transcends to the family and friends of these patients. The challenges that arise from a TBI require additional support from caregivers towards assistance with activities of daily living (ADLs) and instrumental activities of daily living (IADLs). It is common for informal family caregivers to take on this role, which can lead to significant adjustments and changes in family dynamics [7]. Caregivers have reported feeling unprepared and unsupported as they are missing the necessary education and insight to care for an individual affected by TBI at home. As a result, informal caregivers are at risk of also experiencing poor psychological well-being. Studies indicate that up to three-quarters of TBI caregivers suffer from anxiety and depression. Supporting someone with a TBI can be especially challenging for caregivers, as communication barriers – which are often worsened by fatigue – can make it difficult to understand the individual's needs [8]. This can lead to a negative cycle as caregivers lack insight into how the person is feeling, while the individual – affected by comorbidities like fatigue and mood changes – may withdraw further. As a result, they may miss out on the timely care and support essential to their rehabilitation.

Current solutions for supporting social communication between TBI patients and their care teams are limited as many solutions are not focused on specifically communicating fatigue and mood in low effort methods. Reliance upon outdated paper diaries or sporadic self-reporting tools, fail to capture real-time fatigue and emotional changes. When examining available digital solutions, there are available mood tracking tools such as PaceMyDay that help users plan their day based on their energy level, however, NeuroLink is specifically designed with the unique needs of individuals with TBI in mind. It not only tracks fatigue but also captures real-time mood fluctuations and emotional regulation – two critical, yet often overlooked, aspects of TBI recovery. What truly sets NeuroLink apart is its ability to enable secure data sharing. Additionally, using AI-powered pattern recognition, the app transforms subjective experiences into actionable insights that patients and their care teams can use to facilitate communication or adjust support based on how the patient is feeling. These insights can also help care teams recognize early warning signs and personalize treatment.

Implementation of problem-focused coping strategies, designed to support both the individual with TBI and their caregivers, creates a channel to better understand and manage emotional changes as a result of their TBI. Problem-focused coping involves taking practical steps to address a challenge rather than avoiding it or reacting emotionally. Studies have found that this form of communication is associated with higher levels of effective communication and lower levels of caregiver negativity [9]. A mHealth app has the potential to act as a problem-focused coping tool as this platform can offer a structured, real-time method of tracking fatigue and emotional states, which can give caregivers tangible insight into how their loved one is feeling. By increasing awareness and visibility of emotional fluctuations, the app enables more informed, empathetic, and effective communication between the caregiver and the individual with TBI. This can ultimately reduce frustration and misunderstanding, helping caregivers and healthcare providers respond supportively, leading to lower caregiver negativity and better overall interactions, as the research suggests. Given today's increasingly digital landscape, mHealth solutions have seen widespread adoption across various health conditions. These tools offer patients a convenient way to manage their health using devices they already own and are comfortable navigating. Individuals with similar brain injuries have reported positive experiences using self-report features within mHealth apps to track symptoms like fatigue. Real-time monitoring has been associated with a better understanding of one's condition and more effective self-management strategies, further minimizing cognitive bias and supporting a more accurate, data-driven understanding of fatigue [10].

Building on prior research and User Needs Assessments conducted by the current research team – with input from TBI patients, physiotherapists, occupational therapists, nurses, surgeons, and researchers – NeuroLink, an mHealth solution designed specifically for individuals with TBI and their care teams, was developed. NeuroLink empowers users through a simple, intuitive interface that captures real-time insights into fatigue and emotional well-being. The app's design was further strengthened through iterative rounds of prototyping, heuristic evaluations, and usability testing to ensure ease of use and functionality. NeuroLink's primary goal is to address communication barriers caused by fatigue. It reduces uncertainty for caregivers and healthcare providers while ensuring that emotional well-being is integrated into the recovery process. NeuroLink fills a critical gap in the market by offering a seamless, patient-centred, and data-driven tool tailored to the unique challenges of TBI. It relieves the pressure to verbalize complex emotions by enabling silent yet powerful communication, helping patients reconnect with loved ones and care teams. Key features include real-time mood and fatigue logging, AI-powered pattern recognition, and secure data sharing – all designed to turn invisible struggles into actionable insights. By providing structured, personalized data, NeuroLink helps clinicians detect early changes, optimize treatment plans, and ultimately improve recovery outcomes.

2 Development of Prototype Version 1

The first version (v1) of the NeuroLink prototype was developed to directly respond to the user needs identified during the User Needs Assessment, which involved interviews and surveys with key stakeholders such as individuals with moderate TBI, healthcare professionals, and researchers. From these discussions, it became clear that communication barriers, fatigue, and cognitive overload were recurring challenges for TBI patients. Many struggled to express how they felt, especially during moments of low energy, and caregivers were often left guessing how to respond or provide support. Tom, our user persona, reflected many of these realities. He experiences emotional dysregulation, mental fatigue affecting his ability to carry out ADLs, and difficulty articulating how he feels in real-time. Based on this, our design goals focused on creating a solution that was simple, intuitive, and supportive of low-effort communication. The research team aimed to reduce the cognitive burden of interactions, ensure clear feedback, and allow real-time sharing with caregivers to promote better understanding and timely support. NeuroLink prototype v1 was therefore built to support

Tom's core task: communicating his current fatigue level and sending that information to his caregiver with minimal interaction. The communication flow was designed to require no more than six clicks from start to completion, with each screen focused on accessibility and ease of use.

Key features implemented in NeuroLink prototype v1 include:

- A login screen for secure and personalized access
- A personalized welcome screen with Tom's profile and a central button labelled "Start Communicating"
- A home screen with large, clearly labelled buttons for key needs (Fatigue, Food, Noise, Help, Schedule), designed in a tabular format with icons and tooltips
- A fatigue input screen featuring a visual sliding scale for Tom to report his fatigue level, from "Not Tired" to "Exhausted"
- A loading screen indicating the system is processing the input
- An AI-powered prediction dashboard presenting Tom's current fatigue status and expected mood patterns, along with personalized suggestions
- A results-sharing screen allowing Tom to notify his caregiver about his current state
- A confirmation screen that displays after sharing is complete

The primary interaction was carefully designed to accommodate Tom's cognitive limitations, using minimal text, large tap targets, visual indicators, and sequential screen flow. This allowed him to perform a complete communication cycle from opening the app to sharing results without needing to type or verbally express himself.

NeuroLink prototype v1 served as a foundational version, created with both user empathy and functional clarity in mind. It reflected our team's commitment to translating real-world user needs into a digital tool that supports independence, reduces communication frustration, and strengthens caregiver relationships.

3 Heuristic Evaluation Methods

To assess the usability and functionality of NeuroLink v1, our team conducted a structured heuristic evaluation. The goal of this evaluation was to identify any challenges that might limit the effectiveness, efficiency, or satisfaction of the app for its primary users, which include individuals with moderate TBI and their caregivers.

The heuristic evaluation was carried out through a structured process. A total of six evaluators participated, comprising graduate students with academic backgrounds and experience in health informatics, human-computer interaction (HCI), and clinical care. Their domain knowledge ensured that the feedback captured both technical and user-centred design perspectives, particularly in relation to assistive technologies for individuals with cognitive impairments.

Each of the six evaluators independently explored NeuroLink v1 and engaged with a series of key user tasks, simulating the interaction flow of Tom. These tasks included logging into the app, navigating the home screen, selecting and reporting fatigue symptoms, reviewing the AI-generated prediction dashboard, and sharing results with a caregiver. Evaluators were instructed to consider the clarity of the wording used, accessibility, and intuitiveness of each screen and interaction as they progressed through the prototype.

The evaluation process was structured around Nielsen's 10 usability heuristics, which provided a standard framework for identifying and categorizing usability issues. These heuristics are:

(1) Visibility of system status
(2) Match between system and the real world
(3) User control and freedom
(4) Consistency and standards
(5) Error prevention
(6) Recognition rather than recall
(7) Flexibility and efficiency of use
(8) Aesthetic and minimalist design
(9) Help users recognize, diagnose, and recover from errors
(10) Help and documentation

As evaluators completed their walkthroughs of the prototype, they documented any usability issues they encountered. For each issue, evaluators identified which heuristic was violated and provided a written explanation along with recommendations for improvement. This feedback was recorded in qualitative form.

4 Heuristic Evaluation Results

4.1 Prioritization of Heuristic Evaluation Findings

The heuristic evaluation of NeuroLink's prototype v1 provided critical insights that guided the prioritization of usability issues and subsequent refinements in version 2. The research team systematically classified identified issues into four distinct categories: catastrophic, major, minor, and cosmetic. This classification was based on three key criteria: the frequency of mention across independent evaluators, the impact on core user tasks, and the persistence of the issue during interactions.

This prioritization method ensured that efforts were allocated to address the most severe usability barriers first, before optimizing secondary features. Issues reported by multiple evaluators were automatically increased in priority as they indicated systemic usability challenges. For example, the absence of undo functionality and back navigation was noted by four evaluators, the highest frequency of any issue, leading to its classification as catastrophic. Similarly, the lack of a preview screen before data sharing, flagged by three evaluators, was labelled as a major issue requiring immediate resolution.

4.2 Implementation of Changes in Prototype Version 2

The evaluation findings directly informed the specific enhancements introduced in NeuroLink prototype version 2 (v2). A summary of the changes are shown in Table 1. Among the most critical catastrophic issues was the lack of undo and back functionality, which evaluators highlighted as a significant risk for irreversible errors when sharing sensitive health data. To address this, NeuroLink prototype v2 incorporated both an undo option after submission and an ongoing back button throughout the app interface. Another high-priority concern was the absence of a preview screen before sharing results, which increased the likelihood of users unknowingly sharing inaccurate information. In response, Version 2 introduced a confirmation screen, enabling users to review results

Severity	Issue
Catastrophic	Back and Undo Functionality Preview Before Sending to Caregiver
Major	Wording Changes
Minor	Button Colour Consistency Icon Changes "Coming Soon" Pages Contacts Page
Cosmetics	Alignment Fixes

Table 1: Ranking Severity of Issues

before finalizing data sharing, improving the quality of data being shared with caregivers.

Several major issues stemmed from unclear terminology and ambiguous interface elements. For instance, the original "Start Communicating" button misled evaluators, who expected a chat feature rather than symptom logging. This was revised to "Chat Now" to better align with user expectations. Similarly, the vague fatigue-logging prompt, "How do you feel now?" was refined to the more precise "How tired do you feel?", accompanied by a clearly labelled slider. Additionally, AI-generated outputs were clarified, with "Predicted Mood Trends" replaced by the more transparent "Expected Mood Patterns". These adjustments reduced cognitive load and improved the intuitiveness of user interactions.

Minor usability refinements included standardizing button styles and placements to eliminate inconsistencies that caused momentary hesitation. Previously unexplained placeholder features were updated with "Coming Soon" notices to manage user expectations. Visual refinements were also made, such as correcting icon alignment to enhance the app's professional aesthetic. Further enhancements included the introduction of a contacts management page to provide users with the freedom to decide who they will share their results with and the consistent placement of confirmation buttons at the bottom of screens to streamline navigation.

5 Usability Testing Methods

Following the completion of the heuristic evaluation and the development of NeuroLink prototype v2, usability testing was conducted to validate the effectiveness of the improvements. This section outlines the objectives, protocols, and tools employed in the usability testing of NeuroLink.

5.1 Participants

Usability testing was conducted with six participants recruited through word-of-mouth from the research team's personal networks, consisting of friends and family members. A standardized invitation letter was provided to each participant to ensure a shared understanding of the purpose and structure of testing. Participants were eligible if they were between 21 and 65 years of age, English-speaking, and had experience using smartphones and mobile apps. While not mandatory, preference was given to individuals with experience working with TBI patients, as their feedback was expected to better reflect the needs and challenges of the target population. The recruited participants ranged in age from 18 to 54, with four identifying as female and two as male. All held either a bachelor's or master's degree and self-identified their technology experience level as either "advanced" or "expert." "Advanced" users were defined as individuals who regularly use multiple apps and digital tools for work or personal use, while "expert" users were described as those capable of troubleshooting issues and quickly adapting to new technologies. A detailed summary of participant demographics is provided in Table 2.

The demographic factors – namely age, gender, education level, and technology experience – were collected as they can all influence interaction with digital tools. Age can affect motor skills, cognitive processing, and comfort with technology [11,12]. Gender may influence intention to use digital health apps [13]. Education level is associated with health and digital literacy, which can impact how participants navigate an app and interpret its content [14]. Lastly, technology experience is a key factor, as participants with greater technology familiarity are likely to encounter fewer challenges during testing. Collecting this information helped contextualize participant feedback and determine whether usability issues might be attributed to participant characteristics rather than the design of the prototype itself.

Participant ID	Age Range	Gender	Highest Level of Education	Experience with Technology (Smartphones & Apps)
P1	25–34	Male	Bachelor's degree	Expert
P2	18–24	Female	Master's degree	Expert
P3	45–54	Female	Bachelor's degree	Advanced
P4	25–34	Female	Master's degree	Expert
P5	25–34	Male	Master's degree	Advanced
P6	35–44	Female	Bachelor's degree	Advanced

Table 2: Participant Demographics

5.2 Testing Protocol

Each testing session followed a structured protocol. All sessions were conducted one-on-one (one participant and one research team member) and took place in-person at a mutually agreed-upon location, such as a home or local library. Sessions were expected to last approximately 20 minutes, with 10 minutes allocated for the consent process, task explanation, and task completion, and the remaining 10 minutes for the post-test questionnaire and additional feedback.

The consent process involved reviewing and signing a form that outlined the study's purpose and design, potential risks and benefits, data collection and handling procedures, and the voluntary nature of participation, including the right to withdraw at any time. As part of the task explanation, the research team member introduced the think-aloud protocol, which participants were instructed to follow during testing. For those unfamiliar with the method, a brief live demonstration or a YouTube video example was provided. Participants were then presented with the user persona, scenario, and task instructions. In the scenario, they were asked to imagine themselves as "Tom", a 30-year-old patient with a moderate TBI who was feeling fatigued and wished to notify his caregiver, Christina, before her scheduled visit. The task involved logging into Tom's NeuroLink account, recording a fatigue symptom entry, sharing it with Christina, and confirming that the information had

been sent. Participants completed the task using either their own smartphone or a device provided by the researcher. During the session, the research team member used an electronic observation sheet to document task completion, errors, and notable behaviours or comments. Each session was also screen- and audio-recorded to ensure the accuracy of observations. All recordings were securely stored on a restricted-access Google Drive accessible only to the research team and were deleted following analysis to ensure data privacy.

5.3 Data Collection and Analysis

Usability testing for NeuroLink focused on three key metrics: task success rate, ease of use, and usefulness. Each is detailed in the subsections below. Given that individuals with TBI often experience cognitive and emotional challenges [15], it was important to evaluate whether the app was not only functional, but also easy to use and genuinely beneficial for users. These three metrics were therefore selected to guide the testing process.

5.3.1 Task Success Rate: Task success rate was defined as whether participants could successfully use NeuroLink to log their fatigue levels and share the results with their caregiver without making errors or requiring assistance from the research team. This metric was evaluated through direct observation during the testing sessions, including the number of participants who completed the task successfully, as well as notes on any errors, difficulties, or instances in which researcher support was required.

5.3.2 Ease of Use: Ease of use was defined as the extent to which participants could navigate NeuroLink and complete the task with minimal effort. This metric was assessed through the post-test questionnaire (Appendix), specifically the NASA Task Load Index (TLX) – a widely used and validated instrument for measuring perceived workload in healthcare [16-18]. It derives an overall workload score based on participant ratings across six dimensions, including mental demand, physical demand, temporal demand, effort, performance, and frustration [16]. Scores range from 0 to 100, with higher scores indicating greater perceived workload. Although the score was originally designed to be calculated using a weighted average, this approach is often omitted by researchers to streamline administration and analysis [19,20]. Instead, an unweighted average – calculated as the mean of the six dimension scores – is commonly used [19,20]. Several studies comparing the weighted and unweighted approaches have reported mixed findings regarding sensitivity, suggesting that both methods are acceptable [19,20]. Accordingly, this study used the unweighted average to calculate NASA TLX scores.

5.3.3 Usefulness: Usefulness was defined as the extent to which participants perceived NeuroLink to be effective in supporting communication with caregivers, promoting autonomy, and facilitating the management of daily activities. This metric was assessed using five items included in the post-test questionnaire, adapted from the USE Questionnaire developed by Lund [2001] [21]. Items were selected to ensure relevance for TBI patients, while also maintaining brevity of the overall post-test questionnaire to minimize cognitive burden. Each item was rated on a 7-point Likert scale ranging from 1 ("completely disagree") to 7 ("completely agree"), with higher scores indicating greater perceived usefulness. To examine specific aspects of perceived usefulness, the mean and standard deviation were calculated for each item individually.

5.3.4 Thematic Analysis: In addition to the usability metrics, thematic analysis was conducted to gain a deeper understanding of the user experience with NeuroLink prototype v2. A mixed approach to thematic analysis was employed, drawing primarily on deductive methods guided by predefined areas of interest, such as areas of confusion, successful interactions, and user recommendations. The analysis followed Braun and Clarke's six-step framework to examine qualitative data collected during and after the testing sessions [22]. These open-ended insights were documented by the researcher on the observation sheet. The process began with familiarization, during which the observation sheets were thoroughly reviewed. Initial coding was conducted using NVivo software to identify recurring concepts. A reflexive approach was applied, with a single coder iteratively engaging with the data to interpret meanings beyond surface-level responses. Through several rounds of refinement, themes were reviewed and named to accurately represent users' core experiences. The final thematic insights were synthesized with the quantitative findings to provide a comprehensive understanding of participants' challenges, suggestions, and overall experiences using NeuroLink – specifically in relation to logging fatigue symptoms and sending updates to their assigned caregiver, Christina.

6 Usability Testing Results

6.1 Thematic Analysis

Thematic analysis of post-test discussions alongside the observational notes revealed three major themes: Behavioural Patterns, Positive Feedback, and Desire for More Control. These themes reflect participants' interactions with NeuroLink prototype v2, highlighting both areas of strength and opportunities for improvement in user experience.

6.1.1 Behavioural Patterns: Several behavioural patterns emerged during usability testing, particularly in relation to navigation and input elements. Across participants, there was consistent hesitation or confusion around the "Chat Now" button. While all participants ultimately used it to proceed, many assumed it would open a live chat feature rather than lead to symptom logging. Others thought it would lead to another page that would open up some sort of navigation page. Some even considered exploring other tabs, such as "Settings," before being cued or reassured to continue.

"I wasn't sure what Chat Now meant – thought maybe I'd be messaging someone live, not logging symptoms." - P1

"Hmm, it says Chat Now but that doesn't look like where I'd go… but I don't know where else to click." - P3

Another notable behavioural friction point was the fatigue slider, used to report fatigue severity. Participants expressed uncertainty regarding the meaning of the slider's orientation (i.e., which end represented higher or lower fatigue) and commented on the lack of clear feedback on the value being selected. Several users noted it was not intuitive, and suggested adding labels such as "low," "medium,"

and "high" fatigue to better guide them. Others recommended including numerical values to help quantify their symptoms more clearly. One participant even attempted to use the needle-style slider that appeared on the screen but found it non-functional and confusing, noting that it didn't move and lacked visual feedback. These issues indicate that users were not confident in how their input was being interpreted or displayed.

"I'm not really sure what this slider means… like, is the right side more tired? How do I know where to put it? A dropdown with options like 'not tired' or 'very tired' might be easier." - P1

"I couldn't tell which side meant what. Like, is this low or high fatigue? It'd be way easier with numbers or labels… and that little needle didn't even move, I could not get it to work [without guidance]." - P6

Participants also experienced hesitation during the AI insights review phase, particularly regarding whether the feedback reflected their most recent symptom input. Some were unsure how the graphs or recommendations were generated and spent additional time trying to make sense of the information. There was also confusion over how to interpret graphs without clear labels or explanations – especially in cases involving predicted mood patterns.

In the result sharing stage, a notable behaviour was the tendency to add a new contact instead of selecting an existing one (e.g., the caregiver, Christina). In some cases, users attempted to share results without realizing they needed to select the correct contact first. This added unnecessary steps and reflected uncertainty around the workflow for sharing results. Some users expressed confusion about whether they had successfully sent the data or whether the recipient had seen it – suggesting the need for clearer confirmation messages and a more intuitive contact selection interface.

6.1.2 Positive Feedback: Despite the identified usability challenges, participants consistently highlighted several features that they found intuitive, helpful, and user-friendly. One widely appreciated feature was the ability to reverse actions or return to previous screens, which helped reduce the pressure of making mistakes. For example, participants noted that the undo button on the result-sharing screen provided a reassuring safety net, especially when they realized they had not yet confirmed the correct contact. As one P3 put it,

"I liked that I could just go back if I messed something up or I sent something by accident."

The app's clean and minimal interface was positively received by participants, all of whom identified as expert or advanced app users. While they were generally confident navigating digital platforms, they noted that NeuroLink's interface effectively supported efficient task completion without unnecessary roadblocks. Key functions – such as logging in, selecting symptoms, and sharing results – were carried out smoothly. As P4 shared,

"It was pretty intuitive to use, didn't really run into any errors."

While no interface complexity would have likely overwhelmed this user group, the app's ability to deliver a streamlined experience without overcomplicating the interface was seen as a design strength.

Another element that received positive attention was the app's personalized AI insights. While some participants noted that the insights were initially overwhelming or lacked contextual clarity, the overall concept of receiving tailored feedback based on symptom inputs was valued. P4 described the feature as "interesting – it actually told me something based on what was entered." Another participant appreciated that the preview screen before sharing results helped them better understand what the AI-generated information meant and enjoyed knowing feedback could be provided to users to manage their fatigue independently. Participants also recognized the ability of the AI insights to provide personalized feedback, insights and predictions based on data that is inputted by the user overtime.

Several users noted that while they had questions about how mood was being predicted or what the graphs represented, they still appreciated the app's attempt to synthesize their data into actionable insights. One suggestion included having a button to view deeper explanations of each AI output and to include on how to improve mood as well as fatigue as the two can be interrelated, while others requested trend graphs showing which previous entries are used and to help them reflect on changes in fatigue over time.

Overall, these positive interactions show that participants were not only able to complete key tasks, but also saw value in the app's core features – particularly those that supported easy navigation, action reversibility, and personalized feedback.

6.1.3 Desire for More Control: A strong theme across participant feedback was the desire for more customization and system feedback. While participants successfully completed core tasks such as logging fatigue and sharing results, many expressed a need for more granular control over how they described their symptoms and when their data was viewed. Commonly, users emphasized the importance of being able to add additional explanatory notes alongside their fatigue entries – for example, to specify the type of fatigue (e.g., physical or mental) or the specific daily activities they were struggling with (e.g., if they needed someone to do grocery shopping with them, or plan and make a meal). As P4 put it, "It'd be helpful if I could say why I'm tired – like if it's mental or physical fatigue." P5 noted, "I feel like I might need help with a specific task, like cooking instead of cleaning but my caregiver might choose to clean if I don't tell them what I need help with. So, I wish I could add that information." This added context was seen as essential for enabling caregivers to provide more personalized and effective support.

Participants also wanted clearer feedback from the system after submitting information. Several noted they would like to see a notification or confirmation that their caregiver had received and viewed their fatigue report. "I'd like to know if Christina actually saw it. Right now, I think I just sent it but not sure what happens with it," P1 noted. Another suggested the addition of a read receipt or system message confirming that the results were successfully sent and opened.

Users also expressed interest in being able to rate the AI-generated insights, both to confirm accuracy and to help the system improve with future feedback. "It'd be nice if I could give a thumbs up or down on the AI's suggestion, similar to what you do on ChatGPT," P1 noted. This reflects a desire not just for passive information delivery, but for an interactive and learning system that evolves with user input.

Participants also expressed interest in sharing their data with multiple recipients, including family members or caregivers. This highlights a broader vision of the app as a communication hub, not just a symptom tracker. The desire for this kind of flexibility reflects users' expectations that digital tools should support shared decision-making and continuity of care, especially in home or community care settings.

6.2 Task Success Rate

To evaluate usability, the research team observed participants as they completed five core subtasks involved in logging and sharing fatigue levels using NeuroLink. These subtasks were: logging into the app, navigating to the symptom entry screen via the "Chat Now" button, logging fatigue symptoms using a slider interface, reviewing AI-generated results, and sharing the results with a designated caregiver, Christina. Task success, assistance required, and notable errors were recorded and are summarized in Table 3.

(1) *Logging into the App*
All six participants successfully logged into the app without requiring any assistance. This task was completed smoothly, indicating that the login interface was familiar and aligned with users' expectations based on other mobile apps. No errors or confusion were noted during this step.

(2) *Navigating Using the "Chat Now" Button*
Only half of the participants (3/6) immediately identified the "Chat Now" button as the correct next step after login. The remaining three participants either hesitated or clicked on incorrect sections of the interface, such as "Settings." This hesitation stemmed from confusion around the label "Chat Now", which was commonly interpreted as initiating a messaging feature, rather than navigating to symptom selection. These findings suggest the label lacks clarity and may benefit from terminology better aligned with the intended function (e.g., "Log Symptoms").

(3) *Logging Fatigue Symptoms*
Four participants (4/6) completed the fatigue logging task without assistance. However, two participants initially interacted with a non-functional colour slider located above the usable slider and expressed uncertainty about how the functional slider worked, specifically regarding which direction represented higher vs. lower fatigue. This suggests a need for improved real-time feedback, such as labelled endpoints, or dynamic labels that appear as users adjust the slider. The task was ultimately completed by all users, but not without clarification or prompting for some.

(4) *Reviewing Results*
Five participants (5/6) successfully reviewed their results and understood the AI-generated insights. These insights were generally interpreted as helpful and meaningful. However, one participant expressed confusion about the phrase "Expected Mood Patterns", questioning both its meaning and the source of the data used to generate it (such as if the data just inputted was considered and if the data over a few weeks or months is considered). This highlights the importance of making data visualizations and terminology more evident for the users, especially when referencing predicted states or algorithm-generated feedback.

(5) *Sharing Results*
Four participants (4/6) completed the result-sharing task without error, successfully sending their report to Christina. The remaining two participants attempted to add a new contact, unaware that Christina was already available as a preloaded recipient. This indicates some ambiguity in the interface regarding contact selection and suggests that improved contact list visibility may help users complete this task with more confidence.

Task	**Success Rate**	**Assistance**	**Error Notes**
Logging Into App	6/6	0	N/A
Clicking "Chat Now" Button	3/6	3	Clicked on settings; hesitation clicking "chat now" button
Logging Fatigue Symptoms	4/6	2	Tried non-functional slider; needed clarification for max vs no fatigue
Reviewing Results	5/6	1	Confusion over "Expected Mood Patterns" and data source
Sharing Results	4/6	2	Attempted to add new contact

Table 3: Summary of Task Success Rate

6.3 Ease of Use

To assess perceived workload during app interaction, the NASA TLX was administered. As shown in Figure 1, most participants reported low to moderate levels of mental and physical demand, indicating that the core tasks of logging and sharing fatigue data were not cognitively or physically taxing for the majority of users.

However, a few notable spikes emerged across participants, suggesting that some elements of the app may still require refinement. For instance, P5, who reported the highest temporal demand score, was observed hesitating when using the fatigue slider, particularly due to uncertainty around which end represented low or high fatigue, and the lack of immediate feedback while adjusting it. Similarly, Participant 1, who reported elevated levels of frustration, experienced confusion navigating from the home screen, initially unsure whether "Chat Now" would lead to symptom logging or a live chat feature. This user also requested clearer data labels and contextual information when reviewing AI-generated results. These patterns suggest that ambiguous terminology, insufficient feedback from interactive elements, and a lack of transparency in system outputs may have contributed to spikes in perceived effort, time pressure, and user dissatisfaction – even among participants who ultimately completed the tasks.

Each participant's unweighted overall NASA TLX score was also calculated. These scores ranged from 12 to 44 out of 100 (Figure 2), with most participants falling in the low to mid-30s. This reflects a generally low workload, consistent with findings from the task performance and behavioural observation phases.

Overall, these results suggest that NeuroLink was perceived as relatively easy to use, even by participants who encountered minor usability issues. However, spikes in frustration and time pressure indicate that moments of hesitation – such as uncertainty around the "Chat Now" button due to wording, confusion with slider mechanics, or unclear AI-generated feedback – may contribute to unnecessary cognitive strain. Refining terminology, improving interface guidance, and providing real-time feedback may help reduce these issues in future iterations.

Figure 1. Ease of Use (NASA TLX Individual Dimensions)

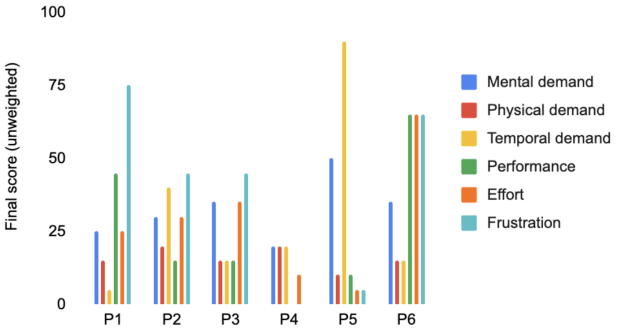
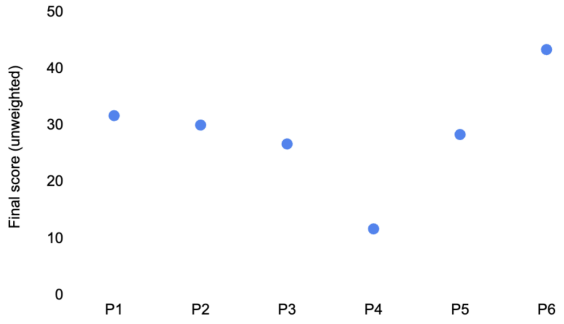


Figure 2. Ease of Use (Unweighted NASA TLX Scores)



6.4 Usefulness

To assess participants' perceptions of the app's usefulness, a 5-item questionnaire with a 7-point Likert scale (1 = completely disagree to 7 = completely agree) was administered. As shown in Table 4, overall perceptions were moderately positive, with average scores across all items ranging from 5.2 to 5.8.

The highest-rated item was "The app is useful" with a mean score of 5.8 (SD = 1.2), suggesting broad agreement on the app's general value. Similarly, "The app does everything I would expect it to do" was rated second highest (mean = 5.6, SD = 0.9), indicating that the app's functions largely met users' expectations.

Items reflecting goal support and autonomy – such as "The app gives me more control over the activities in my life" and "The app makes things I want to accomplish easier to get done" – received slightly lower ratings (mean = 5.2 for both), with higher standard deviations (1.5 and 1.3, respectively). This suggests more variability in responses, indicating that while some users felt empowered and supported by the app, others felt these features could be enhanced. This aligns with qualitative findings from the thematic analysis, where participants expressed a desire for more contextual input, feedback on AI predictions, and the ability to specify or clarify their symptoms before sharing – all of which could improve perceived control and ease in accomplishing health-related tasks.

Importantly, items like "The app meets my needs" and "The app does everything I would expect it to do" had lower SDs (0.9), pointing to a more consistent experience across users for these core functions. This consistency, coupled with relatively high mean scores, suggests that the app's foundational features are functioning reliably, even if certain advanced features (e.g., interactive feedback, notes for caregivers) may still need refinement to enhance user autonomy and task efficiency.

Usefulness Item	Mean (0–7)	SD
"The app is useful."	5.8	1.3
"The app gives me more control over the activities in my life."	5.2	1.6
"The app makes things I want to accomplish easier to get done."	5.2	1.3
"The app meets my needs."	5.4	0.9
"The app does everything I would expect it to do."	5.6	0.9

Table 4: Usefulness Questionnaire

7 Discussion

This study focused on NeuroLink, a mobile health app developed to support individuals with moderate TBIs in tracking their fatigue symptoms and sharing them with caregivers. The primary objective was to determine whether users could effectively navigate the app's core functions, including fatigue logging and sending updates to their designated caregiver. While participants were ultimately able to complete core interactions, the study surfaced several usability concerns affecting confidence, perceived control, and cognitive effort. These challenges mirrored issues found in Schmidt et al.'s [2020] evaluation of their self-guided app for adolescents with mild TBI app, where high usability ratings masked underlying confusion stemming from abstract labels and interface structure. High usability scores can fall short when users experience confusion over interface layout, abstract naming conventions, or unclear module organization. In their case, labels such as "Chill Out!" for relaxation tasks failed to resonate cognitively or emotionally with users, highlighting that technical functionality alone is insufficient for neurodiverse populations [23]. Similarly, NeuroLink's functionally complete interface exposed moments of ambiguity, mental strain, and emotional uncertainty, reinforcing the need to look beyond basic functionality and address intuitive design, clear feedback, and emotional reassurance. The following sections explore five key themes derived from testing and their broader implications for inclusive mobile health design.

7.1 Theme 1: Labeling Confusion and Interface Ambiguity

A recurring concern among participants was the misinterpretation of interface language. The label "Chat Now," for example, was commonly mistaken for initiating a live conversation rather than for launching the fatigue logging function. This led several users to hesitate or reroute to unrelated sections such as "Settings." The fatigue slider presented additional confusion due to the absence of labelled values, directional cues, or feedback to indicate successful input. For individuals with TBI who may experience cognitive fatigue, attention difficulties, and reduced working memory, unclear language and visual ambiguity can significantly increase mental workload. These findings align with Harper et al. [2009], who highlight how visual complexity, such as cluttered layouts, acts as an implicit source of cognitive load, directly impacting how users process and act on information [24]. As such, interface clarity is not cosmetic but fundamental to accessibility, directly influencing a user's decision to continue or disengage, which can impair trust, ease of use, and retention.

7.2 Theme 2: Task Completion Masking User Uncertainty

Although all participants successfully completed key tasks such as logging fatigue and submitting updates, many did so with visible hesitation, reliance on clarification, or trial-and-error behaviours. These observations reveal a critical distinction that completion does not equate to comprehension. Users often expressed uncertainty about whether their input was properly saved or whether their caregiver had received the information. This is especially relevant for TBI populations, where cognitive limitations may obscure recognition of incomplete or incorrect actions. Giakoumis et al. [2014] similarly show that traditional task success metrics can mask more nuanced usability breakdowns, especially among users with cognitive impairments. Their DIAS tool that enables developers to simulate impairments like visual, hearing, physical and cognitive impairments, demonstrates how subtle interaction breakdowns often go undetected in conventional testing, where providing developers with insight over how impairments affect HCI can be considered as treating the issue of accessibility in HCI at its root [25]. This emphasizes the need to assess not just completion, but also user assurance and emotional ease during interaction.

7.3 Theme 3: Positive Response to Undo, Preview, and Personalization Features

Participants consistently expressed appreciation for features that offered an opportunity to preview entries, undo prior inputs, or personalize their submissions. These features were perceived not just as usability aids, but as emotional safeguards that helped users avoid mistakes and feel more in control. For example, users highlighted the ability to select a caregiver as helpful and empowering. These affordances seemed to ease users' concerns, particularly for individuals concerned about clarity in communication or fear of making irreversible errors. Vöcking et al. [2024] affirm the value of these features, emphasizing the role of predictable, user-controlled interactions in fostering psychological safety and adoption of digital health tools. These findings underscore that undo and preview are not convenience features, but critical elements for confidence-building in neurodiverse populations [26].

7.4 Theme 4: AI Insight Confusion and Cognitive Load

The inclusion of AI-generated predictions or predicted mood trends was conceptually well-received by participants, who saw potential value in reflective analytics. However, users found the visual presentation and terminology such as "Expected Mood Patterns" to be abstract, with limited contextual guidance to help interpret what the prediction meant, how it was generated, or how it could be acted upon. These gaps in AI interpretability contributed to confusion and hesitation during the results review task. Chamola et al. [2023] note that necessity for trustworthy AI arises from the increasingly critical roles these systems play in decision-making processes such as healthcare [27]. AI systems lacking transparency can significantly burden users cognitively, especially in healthcare contexts where clarity supports both emotional reassurance and decision-making. For populations managing neurological conditions, ambiguous AI output may be less useful or even counterproductive unless it is framed in simple, explainable, and user-centred terms.

7.5 Theme 5: Lack of Feedback Mechanisms and Visual Cues

Across tasks, participants frequently voiced uncertainty about whether their input had been registered or shared successfully. This was particularly evident with the fatigue slider and share function, which lacked dynamic visual or textual feedback. Some users attempted to repeat the task, while others expressed frustration. These behavioural signals echoed NASA-TLX frustration spikes observed in corresponding tasks. Clear and timely feedback is essential for users with memory or attention limitations. When users do not receive confirmation or reassurance, they may repeat actions unnecessarily, experience frustration, or avoid using the tool altogether. In health contexts, this can compromise the effectiveness of symptom tracking and disrupt caregiver communication. Lam et al. [2018] highlight the critical role of visual feedback in such cases, especially when haptic cues are absent. Their research demonstrates that users perform tasks more accurately and with greater confidence when provided with intuitive visual cues, such as real-time status updates. Visual feedback was shown to be particularly effective for cognitively impaired users, who completed tasks more quickly with visual rather than tactile prompts. The incorporation of grasping visual feedback in their study, produced a more natural interaction [28]. Applied to NeuroLink, these findings suggest that neurodiverse users, particularly those with cognitive strain or uncertainty, benefit from visual confirmation mechanisms that serve both functional and emotional roles. As such, feedback loops must be prioritized not only to ensure system responsiveness but also to support user confidence and trust.

7.6 Limitations

7.6.1 Sample Size and Participant Background: The evaluation involved only six participants, all of whom shared similar educational

and technological backgrounds, such as being classmates or family members of the research team. While this aligns with discount usability testing practices, it poses significant limitations in the context of designing for individuals with TBI, where diversity in cognitive ability, age, and digital literacy is critical. A small, homogenous sample increases vulnerability to dominant opinions and outlier bias, potentially masking usability issues that would arise in more varied populations. Participants' shared academic and technical experiences may have muted detection of more severe interface challenges experienced by users with varying digital literacy, age, or cognitive function. Alroobaea and Mayhew [2014] emphasize that while small samples may uncover some usability issues, they risk missing critical problems and fail to account for individual differences among users. Their empirical findings support the validity of using larger, more diverse participant pools, typically around 16±4 users, to identify over 90 percent of usability problems in complex systems, to ensure inclusive and reliable evaluations, reinforcing the need for broader, representative testing in NeuroLink's future development [29].

7.6.2 Use of Mock Users Instead of Real TBI Patients: Although participants role-played as the user persona, all were adults without lived experience of moderate TBI. This simulation strategy limited the ecological validity of the findings. Critical symptoms such as real-time cognitive fatigue, emotional stress, and task disengagement could not be authentically replicated, which may have reduced the visibility or severity of certain usability issues. This concern is reinforced by the ECOVAL framework developed by Kieffer et al. [2015], which emphasizes that usability evaluations must involve real users, real work goals, and realistic settings to be considered ecologically valid. According to the authors, ecological validity is compromised if the experimental design has not reproduced the user's environment in a representative manner, a shortcoming that can lead to uncertainty about whether or not the study reveals anything about what users actually do in their natural environment. To operationalize this concept, ECOVAL outlines six key dimensions that influence ecological validity: signals, objects, test medium, user interface, task, and behaviour. Signals refer to the sensory inputs that represent the environment, while objects are the physical elements present in that environment. The test medium and user interface relate to the system being evaluated, with the user interface specifically referring to the software. Task describes the nature of the experimental activity performed by participants, and behaviour captures how participants act during the evaluation process. Together, these dimensions provide a structured model for assessing how accurately a usability study reflects real-world use. Without careful consideration of these elements, findings may not only be incomplete but also misleading [30]. Future evaluations of NeuroLink should therefore prioritize the inclusion of individuals with firsthand experience of TBI to ensure meaningful, context-aware results.

7.6.3 Controlled, Task-Based Evaluation: The testing followed a structured sequence of pre-defined tasks, which enabled consistent measurement of workflow but did not account for real-life variability. Users with TBI often interact with mobile health apps during moments of fatigue, distraction, or multitasking. The testing format may therefore overestimate usability by omitting these confounding variables. These concerns are echoed by Ntawanga et al. [2015], who argue that laboratory-based usability evaluations often fail to reflect the realities of mobile app use, as the laboratory does not represent the real environment. To address this limitation, their context-aware model incorporates environmental, technological, and user-specific variables that influence user interaction. The authors define context-awareness as the ability of an app to perceive the user's context and adapt its behaviour accordingly in order to improve the usability of the app. This includes modifying interface elements, system responses, or content in real time based on factors such as the user's cognitive state, location, or environment [31]. For a tool like NeuroLink, which supports users with moderate TBI, usability testing must move beyond static, lab-based scenarios to capture the dynamic, real-world conditions where cognitive fatigue, distraction, and stress often shape user experience.

7.6.4 Subjectivity in Self-Reported Measures: The study's reliance on subjective questionnaires such as NASA-TLX and a modified version of the USE survey provided valuable perception data but may not fully reflect actual usability. Self-reports are vulnerable to mood bias, social desirability effects, and misinterpretation of scales. Although objective metrics have their limitations, they are generally more dependable than self-reports and can offer a clearer picture of system usability [32]. While this study combined subjective and objective methods, including think-aloud protocols, difficulty ratings, post-task surveys, and observational data, to mitigate bias, the influence of self-report limitations cannot be fully eliminated, and may still affect the reliability of the findings. The selection of these metrics should be guided by the system type, time constraints, and the resources available for analysis [32]. Therefore, while NeuroLink's current evaluation incorporates both subjective and objective methods, future studies should continue to strengthen the use of performance-based metrics to reduce reliance on self-reports and enhance the reliability of usability insights.

7.6.5 Incomplete Prototype Functionality: The NeuroLink usability testing was conducted using a partial prototype, which introduced several important limitations. Key features such as expanded symptom tracking and detailed AI explanations, were either non-functional or in early stages, meaning that the testing did not capture the full complexity or scalability of the system. As a result, some usability challenges may not have surfaced simply because critical components were absent. This limitation is reinforced by Salovaara et al. [2017], who introduce the concept of the present-future gap to describe the disconnect between the evaluation context of a prototype and its real-world future use. They explain that a prototype under study exists firmly in the present world, in the circumstances created in the study, but its real context of use is a partially unknown future state of affairs. When studies fail to consider these future contexts, they risk evaluating the wrong futures, contexts, or user groups, potentially leading to false conclusions and expensive design failures. They argue that many prototype evaluations lack projective validity, that is, the ability of findings to hold true in the system's intended future use, emphasizing the need for evaluations that anticipate how user behaviour and context may evolve over time [33]. In the case of NeuroLink, this means the current findings must be interpreted cautiously, as they may not fully reflect

how TBI users will engage with a fully developed, feature-complete version of the app.

7.7 Future Directions

To enhance NeuroLink's utility for individuals with moderate TBI, a multifaceted redesign approach is required. Key recommendations include refining ambiguous labels to reduce interpretive confusion, adding visual anchors and numeric indicators to the fatigue slider, and embedding real-time feedback confirmations for key actions. Clarifying AI outputs through segmented views and explainable summaries will promote better comprehension. Expanding personalization features, including symptom labelling and caregiver note fields, will further align the app with real-world caregiver communication needs.

Future usability testing should involve participants with lived TBI experience and take place in authentic, natural environments. Longitudinal data collection, including in-app analytics and diary-based reflections, will offer insight into sustained engagement and evolving needs. These directions align with Vaghefi and Tulu [2019], who emphasize that continued use of mobile health tools depends on how well the system fits users' needs and supports persistence toward personal health goals. When there is a good fit, users are more likely to sustain engagement, which can lead to improved health outcomes [34]. Adaptive interfaces, simple versus detailed modes, and just-in-time help features should be integrated to reduce cognitive demand and support users with varied levels of digital literacy. These strategies reflect Sridevi's [2014] guidelines for cognitively inclusive user interface design, which emphasize placing the user in control, minimizing short-term memory demands, and maintaining interface consistency. The author highlights that effective interfaces should reduce the need for users to recall previous actions by providing visual cues that support recognition and familiarity. Additionally, information should be disclosed progressively and organized hierarchically, allowing users to access more detailed content only when needed. By hiding technical complexity and presenting tasks at a high level of abstraction, interfaces can lower cognitive burden and better support users with diverse abilities [35]. Furthermore, future iterations must be tested using a fully functional prototype. Participatory co-design, as demonstrated by Joensson et al. [2019], offers a strong model for creating usable and emotionally resonant health technologies by integrating user feedback throughout development. They emphasize that patient satisfaction and positive usability outcomes were directly linked to the integration of end-user ideas, reinforcing the importance of involving and listening to patients to develop appropriate and functional solutions [36].

8 Conclusion

NeuroLink shows strong potential in addressing user needs through simplified input and caregiver connectivity. However, issues with labelling clarity, limited feedback, and unclear AI outputs reveal a gap between user expectations and system behaviour. As Mohamed et al. [2017] note, interfaces must align with users' mental models to ensure usability and trust; when misaligned, users may misinterpret system behaviour or bypass safeguards. They emphasize that intuitive design should reflect protection goals and account for users' tacit reasoning and expectations to prevent critical interaction breakdowns [37].

Effective mobile health design must also address users' emotional and cognitive dimensions. Persada [2018] frames emotional design across three levels: visceral (instinctive, sensory-based reactions), behavioural (conscious interaction with functionality), and reflective (post-interaction interpretation and meaning-making) [38]. To fully support users with cognitive vulnerabilities, NeuroLink must integrate these principles more deeply.

Acharya [2024] cautions that mobile health tools designed without regard for cultural and infrastructural variation risk exclusion. A one-size-fits-all model fails to meet the diverse needs of users in under-resourced or marginalized contexts [39]. NeuroLink's evolution must therefore prioritize inclusive, user-centred, and context-aware design. With continued iterative testing, participatory design, and attention to lived experience, NeuroLink can evolve into a practical and supportive tool, enhancing symptom tracking, strengthening caregiver communication, and promoting everyday independence for individuals managing life after brain injury.

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APPENDIX: USABILITY TESTING QUESTIONNAIRE

Participant ID: _____ Researcher Name: _____ Date: _____

<u>Part I: Demographics</u>

1. **What is your age?**
    a. Under 18
    b. 18-24 years old
    c. 25-34 years old
    d. 35-44 years old
    e. 45-54 years old
    f. 55-64 years old
    g. 65+ years old

2. **What is your gender identity?**
    a. Female
    b. Male
    c. Transgender
    d. Nonbinary
    e. Other (please specify): _____
    f. Prefer not to say

3. **What is your highest level of education?**
    a. Less than high school
    b. High school (including GED)
    c. Some college (no degree)
    d. Associate degree (2-year)
    e. Bachelor's degree (4-year)
    f. Master's degree
    g. Doctoral degree
    h. Professional degree (JD, MD)
    i. Prefer not to say

4. **What is your experience with technology (smartphones and apps)?**
    a. None – I do not use smartphones or apps.
    b. Minimal – I use basic features (e.g., calls, texts, etc.)
    c. Moderate – I use apps, social media, and perform common tasks.
    d. Advanced – I regularly use multiple apps and digital tools for work or personal use.
    e. Expert – I troubleshoot issues and adapt to new technologies quickly.

Part II: NASA Task Load Index (TLX)

Mental Demand - How mentally demanding was the task?

Very Low **Very High**

Physical Demand - How physically demanding was the task?

Very Low **Very High**

Temporal Demand - How hurried or rushed was the pace of the task?

Very Low **Very High**

Performance - How successful were you in accomplishing what you were asked to do?

Perfect **Failure**

Effort - How hard did you have to work to accomplish your level of performance?

Very Low **Very High**

Frustration - How insecure, discouraged, irritated, stressed, and annoyed were you?

Very Low **Very High**

Part III: Usefulness Questions

Instructions: Please rate the following statements based on your experience using the app.

Scale:
1 - Completely disagree
2 - Disagree
3 - Somewhere disagree
4 - Neither agree nor disagree
5 - Somewhat agree
6 - Agree
7 - Completely agree

The app is useful.

1	2	3	4	5	6	7
○	○	○	○	○	○	○

The app gives me more control over the activities in my life.

1	2	3	4	5	6	7
○	○	○	○	○	○	○

The app makes things I want to accomplish easier to get done.

1	2	3	4	5	6	7
○	○	○	○	○	○	○

The app meets my needs.

1	2	3	4	5	6	7
○	○	○	○	○	○	○

The app does everything I would expect it to do.

1	2	3	4	5	6	7
○	○	○	○	○	○	○